Euijin Jung (정의진)

Curriculum Vitae

Ph.D., Brain Robot Augmented INteraction Lab (BRAIN Lab)
Division of Intelligent Robot,
Daegu Gyeongbuk Institute of Science &Technology (DGIST)
R4 711, 333 Techno Jungang-daero, Hyeonpung-myeon,
Dalseong-gun, Daegu, 42988, Republic of Korea
Phone: +82-10-3077-3876
Email: euijin@dgist.ac.kr
Marital Status: Married
Military Service: Completed (Army Sergeant, Active Duty, Jan. 2011 – Oct. 2012)

Specialties

- Artificial Intelligence
- Medical Image Synthesis
- Foundation Models
- Brain Computer Interface (BCI)
- Brain Signal Processing
- Brain MR Processing

Education

Aug. 2023
Ph.D.

Daegu Gyeongbuk Institute of Science & Technology (DGIST)
Department of Robotics and Mechatronics Engineering (Advisor: Sang Hyun Park)
Thesis: *Condition and Modality-guided Medical Image Generation using Deep Learning-based Generative Models*

Feb. 2019
M.S.

Daegu Gyeongbuk Institute of Science & Technology (DGIST)
Robotics Engineering (Advisor: Sang Hyun Park)
Thesis: *Enhancement of Perivascular Spaces using 3D Convolutional Neural Network*

Feb. 2017
B.S.

Korea University of Technology and Education
Electronic Engineering

Research Experience

Daegu Gyeongbuk Institute of Science & Technology (DGIST) (Aug. 2023 – Present)
Brain Robot Augmented INteraction Lab - Professor Jinung An

- Brain signal classification using Image, Audio, and EEG Foundation model [Mar. 2026 – Present]
- Mitigating Subjective Label Bias of EEG Classification [Jan. 2025 – June. 2026]
- EEG and fNIRS Multi-modal Representation Learning [Sep. 2024 – Dec. 2025]
- Pain cortical network visualization using EEG and Fnirs [Aug. 2023 – Dec.2024]

Daegu Gyeongbuk Institute of Science & Technology (DGIST) (Mar. 2017 – Aug. 2023)
Medical Image & Signal Processing Lab - Professor Sang Hyun Park

- Synthetic T2FS-PET generation based on T2FS [Feb. 2023 – April.2024]
- Alzheimer Disease MRI synthesis [Feb. 2019 – Aug.2023]
- Perivascular spaces enhancement in 7T brain MRI [Feb. 2018 – Jan. 2019]
- Knee cartilage segmentation [Jul. 2017 – Jan. 2018]

Project Experience

- Visualization of Pain Cortical Network through EEG/fNIRS data-driven artificial intelligence and its application to pain treatment [Aug.2023 - Present]
- AI-Robot Integrated Digital Twin Care-On Platform for Continuous & Comprehensive Healthcare of the Elderly [Nov.2024 - Present]
- (InnoCORE) Research Group for Bio-Embodied Physical AI [Aug.2025 – Dec.2025]
- Patient-oriented Identification of Pain Chronification Mechanism based on Neural Signal for Development of Non-invasive Treatment [Jan.2025 – Aug.2025]
- Development of Complex Robot's Lifelog Processing and Management based on Indoor-Oriented Heterogenous Robot H/W Platform [Jan.2025 - Aug.2025]
- Classification of Motion Sickness Levels and Prediction of Onset Time Using AI-Based Vehicle User Brain Signals [March.2024 - Dec.2024]
- Explaining AI of current and future state by medical image analysis [Mar.2019 - Aug.2023]
- Oral and Maxillofacial surgery system based on artificial intelligent [Jun.2019 - Aug.2023]
- Diagnosing brain disease based on machine learning with multi-modality image analysis. [Jul.2018 – Aug.2023]
- Brain disease detection based on semi-supervised learning [Jun.2018- Feb.2019]
- Micro medical robot system development for Myocardial infarction [July.2017- Feb.2019]

Awards

- Best Researcher Award in DGIST Robotics and Mechatronics Engineering [2023. Feb]
- Ranked 3rd among 40 teams in Seoul National University Hospital Sleep AI Challenge [2021. Feb]
- 30th Workshop on Image Processing and Image Understanding Best Poster Award [2018. Feb]

Technical Skills

- **Programming:** Experienced in implementing various deep learning models using Python. Proficient in libraries such as OpenCV and Scikit-learn.
- **Deep Learning Frameworks:** Extensive project and research experience with PyTorch and TensorFlow.
- **Operating Systems:** Skilled in deploying deep learning models on Windows, Linux, and macOS.
- **Server Management:** Proficient in using and managing large-scale servers for deep learning training.
- **Brain MRI Preprocessing tools**: Experienced in using FSL and FreeSurfer for brain MRI preprocessing.
- **Brain Signal Preprocessing**: Experienced on EEG data preprocessing using python-MNE library.
- **Version Control:** Maintains a personal GitHub repository for software management [https://github.com/EuijinMisp].

Publications

Journal Publications

1. **Euijin Jung**, Sung Chan Jun, Jinung An, *"EEG-based Multi-Level Pain Classification via Sample Selection to Mitigate Subjective Label Bias", IEEE Transactions on Neural Systems and Rehabilitation Engineering,* vol. 34, pp. 2480-2490, 2026, doi: 10.1109/TNSRE.2026.3692232. **[Impact factor: 5.2, ranking in rehabilitation category: 4th/175]**
2. **Euijin Jung**, Jinung An, "EFRM: A Multimodal EEG–fNIRS Representation-learning Model for few-shot brain-signal classification.", *Computers in Biology and Medicine* 199 (2025): 111292. **[Impact factor: 6.3, ranking in biology category: 7th/107]**
3. **Euijin Jung**, Eunjung Kong, Dongwoo Yu, Heesung Yang, Philip Chicontwe, Sang Hyun Park, Ikchan Jeon, "Generation of synthetic PET/MR fusion images from MR images using a combination of generative adversarial networks and conditional denoising diffusion probabilistic models based on simultaneous 18F-FDG PET/MR image data of pyogenic spondylodiscitis", *The Spine Journal*, 2024. **[Impact factor: 4.9, ranking in clinical neurology category: 35th/277]**
4. **Euijin Jung**, Miguel Luna, Sang Hyun Park, **"**Conditional GAN with 3D discriminator for MRI generation of Alzheimer's disease progression**"**, *Pattern Recognition*, Vol. 133, 2023, 109061, ISSN 0031-3203, https://doi.org/10.1016/j.patcog.2022.109061. [code] **[Impact factor: 7.5, ranking in artificial intelligence: 9th/227]**
5. **Euijin Jung,** Philip ChikonTwe**,** Xiaopeng Zong, Weili Lin, Dinggang Shen, Sang Hyun Park**,** **"**Enhancement of Perivascular Spaces using Densely Connected Deep Convolutional Neural Network", *IEEE Access*, vol. 7, pp.18382-18391, Feb. 2019. [code] **[Impact factor: 3.745, ranking in computer science: 35th/156]**

Selected Conference Proceedings

1. **Euijin Jung**, Hyunmin Lee, Jinung An, "fNIRS Foundation Model for Few-Shot based fNIRS Classification, " *2025 13th International Winter Conference on Brain-Computer Interface (BCI)*, Gangwon, Korea, Republic of, 2025, pp. 1-4
2. **Euijin Jung**, Miguel Luna, Sang Hyun Park, "Conditional GAN with an Attention-based Generator and a 3D Discriminator for 3D Medical Image Generation, " MICCAI, 2021. **[30% acceptance rate]**
3. Won, D., **Jung, E**., An, S., Chikontwe, P., Park, S.H. "Low-Dose CT Denoising Using Pseudo-CT Image Pairs". *PRIME Workshop*, *MICCAI*, 2021.
4. **Euijin Jung**, Miguel Luna, Sang Hyun Park, "Conditional generative adversarial network for predicting 3D medical images affected by alzheimer's diseases," *PRIME Workshop*, *MICCAI*, 2020.
5. **Euijin Jung,** Miguel Luna, Sang Hyun Park, "Conditional Generative Adversarial Network for 3D Medical Image Generation," IPIU 2020.
6. Gyeongmin Lee, **Euijin Jung**, Sang Hyun Park, "Feature Extraction in Optical Coherence Tomography Images for Eye Diseases Using Class Activation Maps," IPIU 2019.
7. **Euijin Jung**, Xiaopeng Zong, Weili Lin, Dinggang Shen, Sang Hyun Park, "Enhancement of Perivascular Spaces using a Very Deep 3D Dense Network", *PRIME Workshop, MICCAI*, 2018
8. **Euijin Jung**, Miguel Luna, Sang Hyun Park, "Knee Cartilage Segmentation Using Semi-Supervised Convolutional Neural Networks," IPIU 2018.
9. Miguel Luna, **Euijin Jung**, Sang Hyun Park, "Automatic Extraction of High-Intensity White Matter Signals in MR Images Using Deep Neural Networks," IPIU 2018.